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PROJECT AND TEAM INFORMATION

Project Title

Smart Public Complaint Analyzer

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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

The Smart Public Complaint Analyzer is a complaint management system which has been designed in order to make the process of registering tracking and managing public complaints more systematic and efficient It enables citizens to submit complaints about common public problems such as damaged roads, electricity failures, water supply issues, sanitation and street lighting. Each complaint is given a unique Complaint ID which citizens can use to monitor its progress The project makes use of C to implement the main data structures and algorithms including a singly linked list for storing the complaints for carrying out search operations, and for managing complaints according to their priority level together with priority-based sorting. C++ is used for the application layer and allows various complaint management functions to be carried out such as updating the status assigning a department and assigning an employee. The C++ application communicates with the C module via the extern C mechanism. The project also has a web-based interface with separate pages for complaint registration, tracking and administration. Browser local storage is used so that the complaint records and the mangement updates can be kept on the website The system is being developed in a step-by-step manner and future improvements will include duplicate complaint detection, complaint statistics and reports, better search and filtering, and further integration. In summary the Smart Public Complaint Analyzer shows how data structures, algorithms, object-oriented programming and web technologies can be put into practical use to create an organised system for managing public complaints.

Updated Project Approach and Architecture

Our project adopts a modular architecture that combines the use of C and C++ for processing and managing complaints, as well as incorporating a web-based interface. The Data Structure Layer is written in C and is responsible for carrying out the basic complaint operations It employs a singly linked list to store the complaint records and provides the facilities to display and search for complaints, to update their status and to assign departments and employees. A priority queue, implemented using linked nodes organises the complaints according to their priority, while prioritybased sorting is used to arrange the complaints from the highest to the lowest priority The Application Layer is developed in C++ It offers a console interface and is in charge of carrying out the various complaint management operations. The C++ application communicates with the C-based data structure module by means of declarations and the extern C mechanism, thus allowing the C functions to be called from within C++ code. The project also includes a web-based interface with separate pages for the home page, complaint registration, tracking and the admin dashboard. Citizens are able to register complaints obtain a unique Complaint ID and track the status of their complaints. Administrators can view the registered complaints, update their status and assign both departments and employees. The website makes use of browser local storage in order to keep the complaint records and the management updates The project is being developed incrementally, with a focus on modular programming, dynamic data structures, priority-based complaint management and the separation of responsibilities. The planned enhancements will be duplicate complaint detection using category, location and text similarity complaint statistics and reports improved search and filtering and further integration and testing of both the console application and the web interface. In general, the architecture illustrates the practical application of C data structures C++ application logic and web technologies in the development of a well-structured public complaint management system

Tasks Completed

Task Completed	Team Member
Complaint registration, linked list implementation, complaint storage, display and search operations	Devraj Rawat
Priority queue, prioriy based sorting, priority handling and planned duplicate complaint detection	Stuti Mahajan
Complaint status updates, department and employee assignment and integration of C modules with application	Rajat
Website development, complaint tracking, admin dasboard, browser based data persistence, and C++ module development	Archi Panwar

Challenges/Roadblocks

One of the main problem we faced in the Smart Public Complaint Analyzer project was integrating the C and C++ code into the same project as the data structure engine was written in C and the application layer was written in C++ so we made sure that functions and data could be properly shared This problem was solved using extern C to make bonding between the two languages and to keep the modules clearly separated. The implementation of priority queue and sorting complains according to their priority was another challenge so we assigned numerical values to high, medium and low priority levels. Managing complaints was one of the main challenge in our project as we had to search complaints, update their status, and assign departmnts and employes properly Also we had to make sure that complaints could be found easily using their unique IDs and that all updates were saved correctly. during development we tested each function separatelly we checked the priority queue sorting and searching and complaint management features using different inputs. We divided the project into smaller modules to make it easier to find and fix errors through this project we gained practical experience in data structures, modular programming connecting different programming languages, and software testing.

Tasks Pending

Task Pending	Team Member (to complete the task)
Develop complaint statistics, reports and improve complaint search and filtering	Devraj Rawat
Implement duplicate complaint detection using category, location and text similarity checks	Stuti Mahajan
Complete C/C++ integration and assist with system debugging and testing	Rajat
Refine website UI and responsiveness and complete website integration with the application	Archie Panwar

Project Outcome/Deliverables

The key result of the project is a complaint management system which helps to organise public complaints from the stage of registration through to tracking and administrative processing Citizens are able to submit complaints, obtain a unique Complaint ID and keep track of the status of their complaints while administrators can see the complaints, update their status and assign both deprtments and employees. The project makes use of fundamental data structures and algorithms written in C, comprising complaint storage based on a linked list searching, a priority queue and priority sorting C++ is employed at the application level for integration with the C modules. A working web interface is also provided for the purposes of registering complaints tracking them and carrying out administration, with the complaint information being stored using browser local storage. The planned deliverables consist of duplicate complaint detection, generation of complaint statistics and reports advanced search and filtering, and further system integration and testing. These improvements will enhance the systems capacity to manage public complaints efficiently.

Progress Overview

Progress Overview The project has successfully put in place its basic architecture and the main complaint management functions. The C module provides facilities for storing complaints, searching, carrying out operations on a priority queue and priority sorting while the C++ application enables the updating of status the assignment of departments and the assignment of employees. The web interface allows for the registration of complaints the generation of a unique ID tracking of complaints and access to an administrative dashboard. Browser-based data persistence has also been implemented so that complaint records and any changes made to the management of the complaints remain available after the page is refreshed. The project is now moving forward to the following stage of development which involves implementing duplicate complaint detection producing complaint statistics and reports, improving search and filtering, and refining the interface. Further integration testing, debugging regular GitHub updates and final documentation are still needed before the project demonstration Generally speaking, the essential complaint management workflow is working although the rest of the work involves carrying out the planned features and verifying the system as a whole.

Codebase Information

Repository link: https://github.com/devrajrawat9027562124-web/Smart-Public$_{C}omplaint_{A}nalyzer$
$Branch : main$
$Importantcommits :$
$InitialprojectsetupandC/C++modulestructure$
$Implementedlinked-list-basedcomplaintstorageandsearching$
$Addedpriorityqueueandpriority-basedsorting$
$Implementedcomplaintstatusupdatesanddepartment/employeeassignment$
$Developedthewebsiteforcomplaintregistration, tracking, andadministration$
$Addedbrowserlocalstoragefordatapersistence.$

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Complaint registration and tracking	Pass	Verified complain registration, unique complaint ID generation and complaint retrieval
Data Structures and Prioritization	Pass	Tested linked list operations, priority queue and priority based sorting
Admin Management	Pass	Verified complaint status updates, department assignment and employee assignment

Deliverables Progress

The core complaint management system has been implemented in C and C++ it includes the use of linked lists for storing complaints searching, priority queue operations, priority sorting, updating the status and assigning the complaints to departments and employees The web interface has also been developed using HTML, CSS and JavaScript, enabling complaint registration, the generation of a unique Complaint ID, tracking, and the provision of an admin dashboard The project is still in progress and further integration testing in C and C++ refinement of the website and regular updates on GitHub are still required. Duplicate complaint detection, complaint statistics and reports advanced search and filtering comprehensive system testing and the final documentation are yet to be carried out These features will be developed and tested before the final project demonstration.